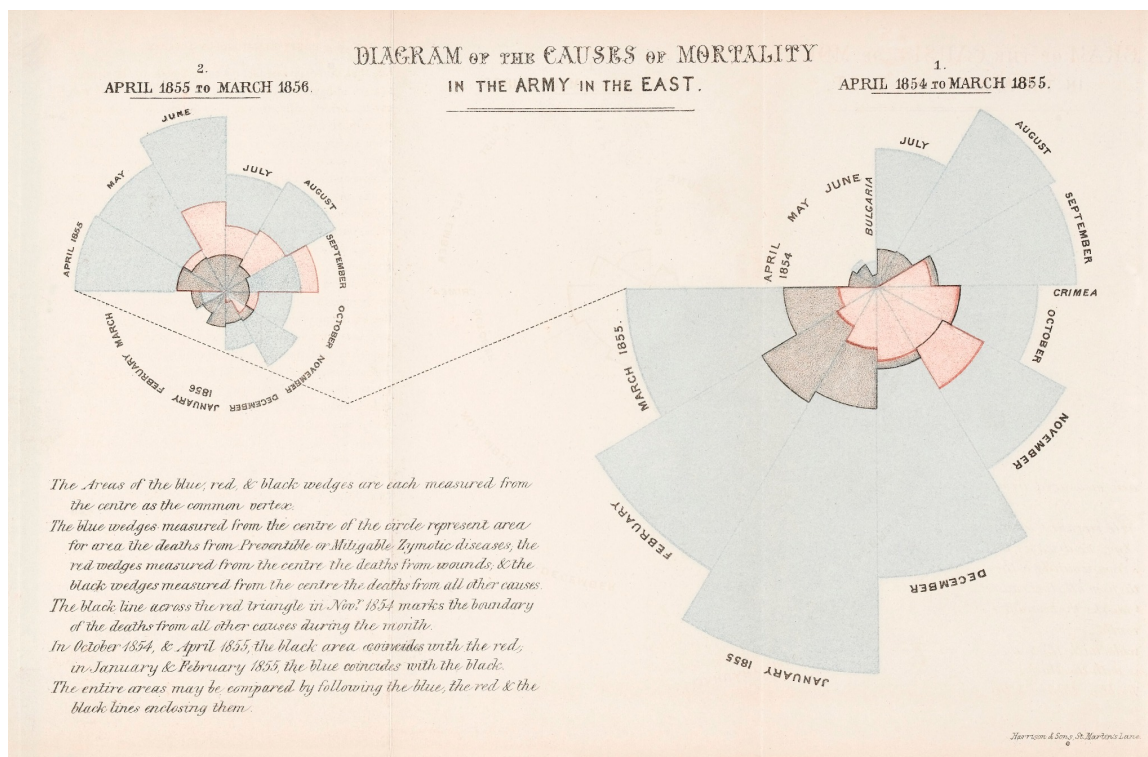


Card 0 | Read the historical instruction

Distribute before the modern candidate; retain the learner's first explanation



“The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.”

What carries the quantity?

Identify the geometric object the instruction names. What does the instruction tell a reader that the silhouette alone does not? Give an explanation rather than a historical claim about who invented visualization.

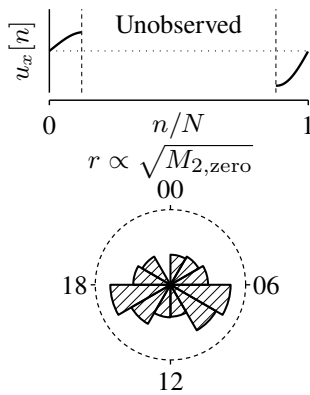
Historical boundary. This source is a reading prompt, not evidence that one chart independently caused reform. The colored areas share a center and overlap; do not read them as a modern additive radial stack. The present case does not reconstruct historical mortality values.

Florence Nightingale (1858), *Notes on Matters Affecting the Health, Efficiency, and Hospital Administration of the British Army*. Source: Wellcome Collection, Public Domain Mark. <https://wellcomecollection.org/works/jxwtstkzc>.

Card 1 | Inspect before endorsement

Distribute without feature definitions, numerical values for the pair, or mask counts

Instructor-constructed



"Area encodes RMS."

Report brief

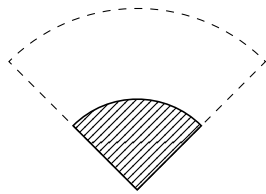
Create a Nightingale-style visualization of this wearable recording and describe its daily pattern.

Candidate report statement

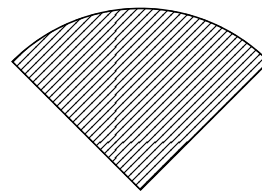
"Area encodes RMS."

The excerpt is a fixed instructor-constructed candidate for an audit, not a transcript from a model. It already indicates an unobserved interval; the exact mask and its treatment arrive later.

State what you think the report represents and which choices remain unresolved. Do not infer the missing values. Then inspect the sector pair below before its values are released.



Sector 1



Sector 2

Prompt. How much larger is the represented quantity in Sector 2 than in Sector 1? Mark the object you used to answer. Is your answer determined by the geometry alone, or does it require an encoding assumption?

The pair has equal angles and a shared reference radius. Values are intentionally withheld at this instructional stage. Its geometry uses the same construction as Fig. 2(a); it is not a second dataset. The complete annotated figures should not be handed out as initial evidence.

Card 2 | Check the mapping

Release after the first ratio judgment and its geometric basis are recorded

The sector pair on Card 1 represents $x = [25, 100]$. Both candidates below use the same positive reference value $x_{\text{ref}} = 100$, the same reference radius $r_{\text{ref}} = 1$, equal angles, and zero inner radius.

Candidate A

```
x = [25 100];
r = sqrt(x / 100);
caption = 'Sector area represents x.';
```

Candidate B

```
x = [25 100];
r = x / 100;
caption = 'Sector area represents x.';
```

Decide. For each candidate, compare the implemented mapping with the caption. Should the candidate be retained, relabeled, or recomputed? What ratio would someone obtain from the radius, and what ratio from area?

Reference resource, if needed. For an equal-angle sector measured from the center, area is proportional to the square of radius. A value intended to be carried by area therefore determines radius through a square root. This is a mapping rule, not a rule that all polar plots must use square-root radii.

Discuss. Can a numerical answer be right for the wrong reason? What evidence in an audit would show which geometric object the reader actually compared?

Review note. Both candidates are instructor-constructed. Their provenance does not determine correctness. Write the decision and its reason before consulting the instructor reference calculations.

Card 3 | Check feature and time support

Release after the geometry audit; do not assume every candidate needs changing

A preprocessed three-axis acceleration sequence has vector-magnitude mean square M_2 and RMS R . Mean square averages the sum of squared axis values; RMS is its square root. In these synthetic examples, acceleration is in arbitrary units, so mean square is in squared arbitrary units. A physically calibrated recording would require its own units and preprocessing record.

Candidate C

A signal and an amplitude-scaled version have RMS values $R = [1, 2]$. The sector angles are equal.

```
R = [1 2];
r = R / 2;
caption = 'Area represents mean-square acceleration.';
```

Decide. Is the linear radius necessarily an error? Give a reason for retaining or revising this candidate.

Candidate report from Card 1

```
% U0 is a fixed preprocessed record with missing positions set to 0.
M2_zero = mean(sum(U0.^2, 2));
r = sqrt(M2_zero / 100);
caption = 'Area encodes RMS.';
```

Decide. Which feature does area actually encode? If the intended feature is mean square, what is the smallest defensible repair? If the intended feature is RMS, what must change instead?

Keep the audit open. A feature-label repair does not settle how absent observations should enter the summary. Record that issue for the next release rather than silently treating a consistent caption as the end of the audit.

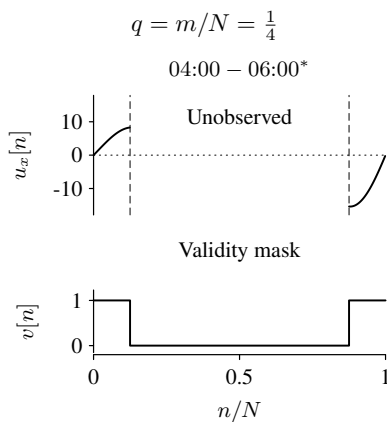
Interpretation. The snippets specify candidate calculations. Their results do not define fatigue, clinical risk, mechanical power, or work performed.

Temporal support prompt. What analysis window produced this statistic? Could the same RMS occur with different burst timing or duration? The optional signal-audit extension develops these questions using a separate 100-Hz synthetic example.

Card 4 | Check observation coverage

Release the counts and fixed validity mask before asking for a final interpretation

(a) Missing observations



The selected schematic interval has **800 sample positions**, of which **200 are valid**. The observed squared sum is **20,000**. The same observations are used in both candidates; only the denominator and interpretation differ.

Candidate D

Divide the squared sum by all 800 positions. Its proposed interpretation is: “The smaller sector indicates greater fatigue.”

Candidate E

Divide by the 200 valid positions, display coverage separately, and restrict the description to the observed-sample statistic.

What changed, and what did not?

Explain the difference between the summaries without invoking two people or two different observed signals. Is the valid-sample mean necessarily the mean of the complete interval? How would a no-observation interval be shown?

Conditions of this teaching comparison. The mask is applied after preprocessing is fixed. The missing entries are not available to the learner. Re-filtering after filling gaps is outside this comparison. The two-hour labels organize schematic bins; 800 points do not establish a physical wearable sampling rate.

Return to the claim. Distinguish what can be said about the observed statistic from what would be required to compare full intervals or infer fatigue. Write the evidence you would request rather than insert assumed values.

Reading note. The visible mask is an excerpt from Fig. 3. Consult the instructor reference calculations after recording a decision; valid-sample normalization does not recover the missing data.

Card 5 | Revise and justify

Use after all three audits; collect the final individual decision record

Complete the reverse audit across graphical encoding, feature semantics, temporal support, and observation support, then justify the report to another reader. It may preserve parts of the candidate, modify others, and withdraw unsupported interpretations. Do not change a correct artifact merely because it was generated or because another candidate was wrong.

Submit three connected artifacts

Code or computational annotation. Identify the statistic, units, time window, duration, mask, denominator, and scale. The explanation must be specific enough for a reviewer to locate the relevant operations.

Annotated visualization. State the geometric carrier, keep comparable scales, and report coverage without confusing missing observations with a zero-valued feature. You may propose another chart type if you explain its suitability; reproducing a rose is not the grading objective.

Bounded interpretation. State what the observations support, what they do not settle, and what extra evidence a stronger claim would require.

Leave a decision trail

Complete the worksheet's final record: earlier statement, newly available evidence, action taken, and remaining uncertainty. Keep any correct initial statement and explain why it remains justified.

For an optional live GenAI run

Retain the model or service name, date, complete prompt, unedited output, checks, and edits. Distinguish the live result from instructor-constructed variants. A second assurance from the model is not an independent verification.

Debrief. What did the historical reading instruction contribute beyond the appearance of the rose? At what point did you have enough evidence to describe a statistic, and at what point would you still need to suspend a statement about a person?